MAIDA 2027 Bridging Program

PYTHON PROGRAMMING ESSENTIALS

Session 4 of 5

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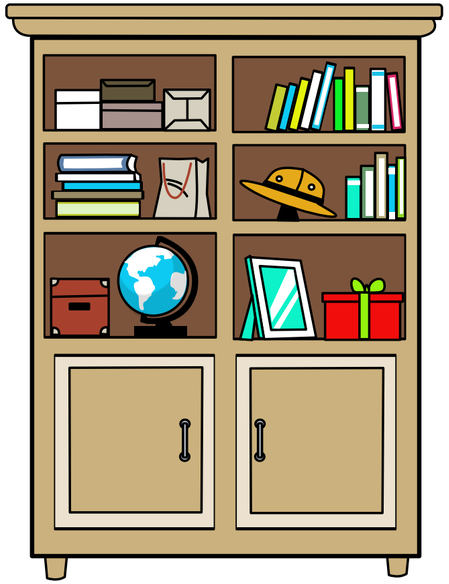
WHAT ARE COLLECTIONS?

Collections are **container** data types that allow you to store groups of various data



WHAT ARE COLLECTIONS?

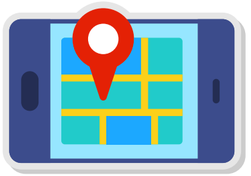
Python has several built-in container data types

List

can store a sequence of values (mutable)

Tuple

can store a sequence of values (immutable)



```
(lat, long)
(14.5826, 120.9787)
```

Dictionary

collection of key-value pairs

Set

unordered unique collection of values

BRIDGING SESSION MINI PROJECT

Create (vibe code) and **present** a simple Python program that reflects your personal interests, creativity or goals as a future AI/Data Analytics practitioner.

Briefly explain:

1. **Program Idea** - What does the program do?
2. **Code Walkthrough** - Explain logic of the code.
3. **Concepts Used** - Identify the Python concepts you applied.
4. **Demo** - Run your program and show it works.

RESPONSIBLE VIBE CODING

Some rules to live by when vibe coding:

1. **Human-in-the-Loop & Review** - Never trust AI blindly.
2. **Small, Atomic Tasks** - Split projects into tiny, manageable tasks to prevent AI confusion and improve accuracy.
3. **Rigorous Testing** - Write test cases for each feature. Tests help describe what “working” looks like
4. **Version Control** - Use Git to create save points of your code.
5. **When Confused, Stop Coding** - You can’t build what you can’t explain.

References:

1. https://www.reddit.com/r/ClaudeCode/comments/1pu2i00/10_rules_for_vibe_coding/
2. Gemini from Google Search